

give me the info for Long Rang UHF RFID reader HX-R01B-9

I'm checking the manufacturer and reseller sources first so I can give you a clean summary of the actual specs, interfaces, and likely use cases.

I found the product family and a matching listing, but the manufacturer pages are loading poorly. I'm cross-checking against reseller pages to avoid giving you guessed specs.

Thought for 46s

Here's the key info I could verify for the **Long Range UHF RFID reader HX-R01B-9**:

What it is

The **HX-R01B-9** is a **UHF integrated RFID reader/writer** in the **HX-R01B** family made by **Shenzhen Haixin IOT Technology Co., Ltd.** The family listing shows it alongside **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, and Haixin describes the series as a high-performance integrated UHF reader for applications such as **vehicle management** and **access control systems**. Haixiniot +2

Confirmed specs I found

From a matching product listing for **HX-R01B-9**:

- **Model:** HX-R01B-9 物流技术网
- **Dimensions:** 260 × 260 × 100 mm 物流技术网
- **Housing material:** engineering plastic 物流技术网
- **Power supply:** DC 9V to 30V, with recommended power capacity **not less than 60W** 物流技术网
- **RF output power:** ≤ 30 dBm 物流技术网
- **Interfaces:** TCP/IP or RS232 selectable, plus RS485, with optional Wiegand 26/34 and other user interfaces 物流技术网
- **Operating mode:** FHSS or fixed frequency, software configurable 物流技术网

Frequency / UHF band

Haixin's related product page for this family references a **wideband UHF design** supporting regions such as **902–928 MHz** and **865–868 MHz**, which is consistent with UHF EPC Gen2 style deployments. Haixiniot

Protocol support

I could not fetch the full official datasheet page directly, but Haixin's nearby UHF product listings show support in this product line for standards including **EPC C1G2 / ISO 18000-6C**, and in some cases **ISO 18000-6B / GB/T 29768-2013**. That suggests the HX-R01B family is intended for standard UHF passive RFID use, but I would treat **the exact protocol list for HX-R01B-9 as needing final confirmation from the official datasheet.** 物流技术网 +1

Typical use cases

Based on the manufacturer's product category and the exposed interfaces, this reader is aimed at:

- vehicle gate / parking access
- long-range card or tag identification
- industrial access control
- asset or personnel identification in outdoor/semi-outdoor setups Haixiniot +1

What the "-9" likely means

Because the same family is sold as **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, the suffix likely corresponds to the **integrated antenna variant**, commonly interpreted as **4 dBi / 9 dBi / 12 dBi** versions. I found strong family evidence for those model groupings, but I did **not** find a directly readable official line explicitly stating "9 = 9 dBi," so that part should be treated as an informed inference rather than a confirmed spec. Haixiniot +1

Practical summary

So, the **HX-R01B-9** appears to be a **fixed, integrated long-range UHF RFID reader** with:

- built-in antenna form factor
- up to **30 dBm** output
- **TCP/IP / RS232 / RS485** connectivity
- optional **Wiegand 26/34**
- support for configurable regional UHF operation
- use in **vehicle management** and **access control** 物流技术网 +2

The main thing I could **not** verify cleanly is the full official datasheet content, because the manufacturer pages were timing out during fetch.

What I found but could not fully confirm

I found the manufacturer and a product-specific listing, but the official detail pages were not readable through the browser tool, so some items like **exact read distance, protection rating, tag capacity, and exact protocol matrix** remain unconfirmed from primary-source text. The strongest verified fields were the **dimensions, supply, output power, interfaces, and family identity**. Haixiniot +2

I can also give you a **datasheet-style summary sheet** for HX-R01B-9, or help compare it with HX-R01B-4 and HX-R01B-12.



give me the info HX-R01B-9 and GARF_RB01

I'm checking both model names side by side so I can separate what is clearly verified from what only appears in marketplace references.

Thought for 17s

Here's a side-by-side summary of what I could verify for **HX-R01B-9** and **GARF_RB01**.

1) HX-R01B-9

What it is

HX-R01B-9 is part of the HX-R01B series from **Shenzhen Haixin IOT Technology Co., Ltd.** The manufacturer's product catalog lists **HX-R01B-4 / HX-R01B-9 / HX-R01B-12** together as **UHF RFID integrated readers**. Haixiniot +1

Verified family-level positioning

Haixin presents this series as a **UHF RFID integrated reader** product line. Haixiniot +1

Previously verified model-specific details

For HX-R01B-9, the model-specific listing I found indicates:

- **Type:** long-range UHF integrated RFID reader/writer
- **Dimensions:** 260 × 260 × 100 mm
- **Housing:** engineering plastic

- **Power input:** DC 9V–30V
- **Recommended power capacity:** $\geq 60\text{W}$
- **RF output power:** $\leq 30\text{ dBm}$
- **Interfaces:** TCP/IP, RS232, RS485
- **Optional interfaces:** Wiegand 26/34
- **Mode:** FHSS or fixed frequency, configurable in software

These fields came from the product-specific listing I referenced earlier for HX-R01B-9.

Haixiniot

Likely usage

The Haixin catalog positions this UHF reader family for applications such as **vehicle management** and **access/access-control style systems**, which fits a long-range fixed UHF reader. Haixiniot

Important note

I could confirm the **HX-R01B** family from the manufacturer catalog, but I could not directly load a clean official datasheet page for **HX-R01B-9** during this search session. So the most solid facts are the family identity plus the model details from the matching listing.

Haixiniot +1

2) GARF_RB01

What I could verify

I could not find a reliable public manufacturer page, datasheet, or catalog entry for **GARF_RB01** in this search. The search results did not return a credible product match for that exact identifier. BidNet Direct +2

What that suggests

Because there is no strong public match, **GARF_RB01** is likely one of these:

- an internal product code
- a custom/private label model name
- a project or assembly identifier
- a document-level part name rather than a market-facing product model

That is an inference based on the absence of trustworthy public hits, not a confirmed manufacturer fact. BidNet Direct +2

3) Most likely relationship between them

Based on what is publicly visible:

- **HX-R01B-9** looks like a **real commercial UHF RFID reader model** from Haixin.
Haixiniot +1
- **GARF_RB01** does **not** currently look like a publicly documented commercial model number. BidNet Direct +2

So GARF_RB01 may be your internal name for a device based on, paired with, or documented around HX-R01B-9, but I could not verify that link publicly.

4) Clean practical takeaway

HX-R01B-9

A **fixed long-range UHF RFID reader**, likely for gate/access/vehicle identification, with Ethernet/serial connectivity and up to **30 dBm** RF output. Haixiniot +1

GARF_RB01

Not publicly verifiable from reliable sources at the moment. It may be a **custom or internal identifier** rather than a vendor model. BidNet Direct +2

5) What I found was insufficient

I found clear catalog evidence for the **HX-R01B family**, but not a readable official datasheet page for **HX-R01B-9**, and I found **no trustworthy public source** that clearly identifies **GARF_RB01**. That means I can describe HX-R01B-9 reasonably well, but I cannot honestly give confirmed specs for GARF_RB01 from public web sources. Haixiniot +3

Send me a photo, label, datasheet, or PCB silkscreen for **GARF_RB01**, and I'll map it against **HX-R01B-9** precisely.



datasheet-style summary sheet for HX-R01B-9

I'm pulling the strongest manufacturer and listing details together and formatting them like a compact datasheet, with uncertain fields clearly marked.

Thought for 9s

HX-R01B-9 — Datasheet-Style Summary Sheet

Product overview

The HX-R01B-9 is a UHF integrated RFID reader/writer in Haixin’s HX-R01B family. The manufacturer groups it with HX-R01B-4 / HX-R01B-9 / HX-R01B-12, and describes the family as a high-performance UHF integrated reader for uses such as vehicle management and access control systems. Haixiniot +1

General description

This reader is an all-in-one fixed UHF reader with integrated RF/antenna structure, intended for long-range tag identification in fixed installations. Haixin states the family uses proprietary high-efficiency signal processing to maintain a high read rate. Haixiniot +1

1. Identification

Item	Value
Model	HX-R01B-9
Product class	UHF integrated RFID reader/writer
Product family	HX-R01B series
Manufacturer	Shenzhen Haixin IOT Technology Co., Ltd.

These points are supported by the manufacturer family catalog and the model-specific mobile product page. Haixiniot +2

2. Physical specifications

Item	Value
Dimensions	260 × 260 × 100 mm

Item	Value
Housing material	Engineering plastic

The dimensions and material are listed on the model-specific product page. [物流技术网](#)

3. Electrical specifications

Item	Value
Input voltage	DC 9V to 30V
Recommended power capacity	Not less than 60W
RF output power	≤ 30 dBm

These electrical figures come from the HX-R01B-9 product listing. [物流技术网](#)

4. Communication interfaces

Item	Value
Primary interfaces	TCP/IP or RS232 selectable
Additional interface	RS485
Optional interface	Wiegand 26/34

The model page states the unit provides **TCP/IP and RS232 as selectable options**, plus **RS485**, with optional **Wiegand 26/34** and other custom interfaces. [物流技术网](#)

5. RF / operating mode

Item	Value
Working mode	FHSS or fixed frequency

Item	Value
Configuration	Software configurable

The model page explicitly lists **FHSS** and **fixed-frequency** operation modes. [物流技术网](#)

6. Frequency band

Haixin’s related HX-R01B family page indicates support for common regional UHF RFID bands such as **902–928 MHz** and **865–868 MHz** for the series. I would treat this as **family-level guidance** unless you have the exact HX-R01B-9 datasheet page for direct confirmation. [Haixiniot](#)

7. Typical applications

The manufacturer describes the HX-R01B family as suitable for:

- **vehicle management**
- **gate / access control**
- **fixed long-range identification deployments**

That fits the interface mix and integrated-reader form factor of the HX-R01B-9. [Haixiniot +1](#)

8. Notes on model suffix “-9”

Because the family is sold as **HX-R01B-4 / HX-R01B-9 / HX-R01B-12**, the suffix likely denotes an **antenna-gain variant** such as **4 dBi / 9 dBi / 12 dBi**. I did **not** find a directly readable official line confirming that on the pages I could access, so this should be treated as an informed inference, not a confirmed specification. [Haixiniot +1](#)

9. Compact engineering summary

HX-R01B-9 appears to be a **fixed UHF RFID integrated reader** with **up to 30 dBm output**, **9–30V DC input**, **Ethernet/serial connectivity**, and optional **Wiegand integration**, aimed at **vehicle and access-control scenarios**. The most solidly verified fields are the **model family**, **dimensions**, **power input**, **RF output**, **interface set**, and **operating modes**.

[Haixiniot +1](#)

10. Unverified or not cleanly confirmed in the accessible sources

I could not cleanly verify these from the sources I accessed:

- exact read distance
- exact tag protocol list for HX-R01B-9 specifically
- IP rating
- operating temperature
- exact confirmation that “-9” = 9 dBi

So those fields should be left blank or marked **TBD** until you have the official datasheet PDF or product manual. Haixiniot +2

Ready-to-copy version

Model: HX-R01B-9

Type: UHF integrated RFID reader/writer

Manufacturer: Shenzhen Haixin IOT Technology Co., Ltd.

Series: HX-R01B

Physical:

- Dimensions: 260 × 260 × 100 mm
- Housing: Engineering plastic

Electrical:

- Supply voltage: DC 9V–30V
- Recommended power capacity: ≥60W
- RF output power: ≤30 dBm

Interfaces:

- TCP/IP or RS232 selectable
- RS485
- Optional Wiegand 26/34

Operation:

- FHSS or fixed-frequency mode

I can also turn this into a **formal one-page datasheet template** with sections like **Features**, **Specifications**, **Interface Definition**, and **Applications**.



Sources

give me the protocol for HX-R01B-9

I'm checking whether Haixin publishes a protocol/manual for this exact reader or whether only the air-interface standards are visible publicly.

Thought for 21s

For **HX-R01B-9**, there are **three different "protocol" layers** people usually mean:

1) Tag air-interface protocol

This is the protocol between the reader and the RFID tags.

The strongest public evidence I found says the **HX-R01B family supports ISO18000-6C/6B dual protocol**. The manufacturer snippet for the related Haixin page states that the UHF access-control/channel system supports "**ISO18000-6C/6B dual protocol**", and that page also includes the **HX-R01B-6 / HX-R01B-9 / HX-R01B-12** family. Haixiniot +1

So the **RFID air protocol** for HX-R01B-9 is most likely:

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B** Haixiniot +1

In UHF RFID terms, **ISO 18000-6C** is the same family commonly referred to as **EPC Class 1 Gen 2 / EPC C1G2**. EM Microelectro... +1

2) Host communication interface protocol

This is how your controller or PC talks to the reader.

For HX-R01B-9, the public model listing indicates these host interfaces:

- **TCP/IP**
- **RS232**
- **RS485**
- **optional Wiegand 26/34** Tesike Electronics

So if you are integrating the device into a system, the reader likely communicates outward through one of those interfaces.

3) Command-set / packet protocol

This is the exact byte-level reader command format over RS232, RS485, or TCP/IP.

I could **not** verify a public manual or official command document for the exact **HX-R01B-9 host command protocol**. So I cannot honestly give you the frame header, checksum, command IDs, or packet examples for this model from verified sources.

Practical answer

If you ask, "What protocol does HX-R01B-9 support?", the safest verified answer is:

- **RF tag protocol: ISO18000-6C / ISO18000-6B** Haixiniot +1
- **Reader-side communication interfaces: TCP/IP, RS232, RS485, optional Wiegand 26/34** Tesike Electronics
- **Exact serial/TCP command protocol manual: not publicly verified from the sources I could access**

If you want, I can make you a **protocol section in datasheet format** or help you infer the **serial command structure** from a photo of the manual, SDK, or test software.



what protocol does HX-R01B-9 support?

I'm verifying the exact standards wording from the vendor-facing pages so I can separate tag protocol support from the reader's host-side interfaces.

Thought for 7s

HX-R01B-9 supports:

RFID tag protocol

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B**

Haixin's product page for the HX-R01B-6 / HX-R01B-9 / HX-R01B-12 family explicitly says the system supports **"ISO18000-6C/6B dual protocol."** Haixiniot

Reader communication interfaces

These are not RFID air protocols, but the ways your host system can connect to the reader:

- TCP/IP
- RS232
- RS485
- optional Wiegand 26/34

These interfaces are listed on reseller pages for the long-range integrated UHF reader variant. Tesike Electronics

What that means in practice

So, if you are asking about the **RFID protocol**, the answer is:

- ISO 18000-6C
- ISO 18000-6B Haixiniot

If you are asking about the **PC/controller connection protocol**, it supports communication over:

- TCP/IP
- RS232
- RS485
- optional Wiegand Tesike Electronics

I could not verify the exact **byte-level command protocol manual** for HX-R01B-9 from a public official source.



give me more detailed description for these protocols

I'm expanding each one separately: first the RFID air protocols, then the host-side interfaces, because they do different jobs in the system.

Thought for 13s

Sure — here is a **more detailed description** of the protocols relevant to **HX-R01B-9**.

1) ISO/IEC 18000-6C

This is the **main UHF passive RFID air-interface protocol** used by many modern UHF tags and readers. Haixin's family page says the HX-R01B line supports **ISO18000-6C/6B dual protocol**. GS1 +1

What it does

ISO/IEC 18000-6C defines how the **reader talks over radio to passive UHF tags**. GS1 describes Gen2 as the air-interface protocol that defines the **physical and logical requirements** for an RFID system of interrogators and tags. It is the protocol commonly known as **EPC Gen2 / EPC Class 1 Gen 2**. GS1 +2

What it is used for

It is widely used for:

- **inventory and logistics**
- **asset tracking**
- **vehicle identification**
- **access and gate systems**
- **supply-chain labeling** GS1 +2

Key characteristics

In practical terms, 6C / Gen2 gives you:

- **anti-collision**, so one reader can identify many tags in the field
- **read and write operations** on supported tag memory
- support for standard UHF regions and deployments
- interoperability with a large ecosystem of Gen2 tags GS1 +1

Why it matters for HX-R01B-9

For a long-range reader like HX-R01B-9, this is likely the **primary protocol used for actual tag reading**. If your tags are normal UHF EPC tags, this is probably the protocol you will use most often. GS1 +1

2) ISO/IEC 18000-6B

This is an **older UHF RFID air-interface protocol**. Haixin states the family supports it alongside 6C. GS1 +1

What it does

Like 6C, 6B defines the **radio communication method between the reader and UHF RFID tags**, but it is a different standard generation. It is typically associated with older or legacy UHF tag deployments. The contrast in modern tag datasheets is telling: many current UHF tags explicitly advertise **ISO/IEC 18000-6C compliance**, not 6B, which reflects where the market standardized. Atlas RFID Store +1

Why support both

If a reader supports both **6B and 6C**, that usually means:

- it can work with **older installed tag systems**
- it can also work with **newer EPC Gen2-style tags**
- it is more flexible in mixed or transitional environments

That is useful in gate control and industrial systems where some deployments still use older tag types. This last point is an inference from how dual-protocol readers are commonly positioned, supported by the vendor's explicit "6C/6B dual protocol" wording.

GS1

3) TCP/IP

This is **not the RFID tag protocol**. It is the **host communication interface** between the reader and a PC, controller, server, or access-control system.

What it means

When the reader says it supports **TCP/IP**, it usually means the unit can connect over **Ethernet** and exchange data with software on a network. The reader may:

- send tag IDs to a server
- receive configuration commands
- report real-time reads to a control system

Haixin/reseller listings for this product family expose **TCP/IP** as one of the available interfaces. HID Global

Typical use

You would use TCP/IP when:

- the reader is mounted at a **gate, parking entrance, or doorway**
- your application runs on a **remote PC or server**
- you want easier integration into a **LAN-based system**

Benefit

TCP/IP is usually the easiest choice when you want the reader connected directly into a software platform or networked controller. That is an integration inference based on the nature of Ethernet-connected RFID readers. The source support here is the availability of the TCP/IP interface itself. HID Global

4) RS-232

This is a **serial host communication standard**, again between the **reader and your host device**, not between the reader and the RFID tag.

What it does

RS-232 is a classic point-to-point serial interface used to connect equipment such as scanners, PLCs, and industrial devices. National Instruments describes RS-232 alongside RS-422/485 as part of standard serial communication. Advantech notes RS-232 is typically used over relatively short distances and is a single-ended interface. NI +2

Typical use with RFID

With HX-R01B-9, RS-232 would typically be used when:

- the reader is connected directly to a **PC**
- the distance is short
- you want simple serial command/control integration
- you are using vendor demo software or a local industrial controller

Practical behavior

In many devices, RS-232 carries a **vendor-specific command set** over a serial COM port. I could not verify the exact byte-level HX-R01B-9 command format publicly, but that is the usual pattern for readers exposing RS-232. The factual part is the interface support; the command-set pattern is a cautious inference. NI +1

5) RS-485

This is also a **serial host communication standard**, but it is usually preferred in **industrial environments**.

What it does

RS-485 uses **differential signaling**, which is better for longer cable runs and noisier environments than RS-232. NI and Advantech describe RS-485 as part of the standard serial family and contrast it with RS-232's single-ended signaling. NI +2

Why it is useful

RS-485 is common when:

- the reader is installed **far from the controller**
- the site has **electrical noise**
- the reader is part of an **industrial or building-control bus**
- you want more robust field wiring than RS-232

Typical RFID role

With a long-range fixed reader, RS-485 is often used for:

- factory or warehouse installations
- barrier/gate controllers
- multi-device control systems

Again, the exact HX-R01B-9 serial command frames were not publicly verified, but RS-485 would typically carry that same vendor command protocol over a more robust physical layer. NI +1

6) Wiegand 26-bit / 34-bit

This is different from TCP/IP and serial links. **Wiegand** is a very common **access-control output format/interface**.

What it does

Wiegand is used to send credential data from a reader to an **access-control panel**. Industry references describe it as a wiring/interface standard common in card readers and access systems, and note that **26-bit** and **34-bit** are among the common output formats.

Safe and Sound ... +2

What 26-bit and 34-bit mean

The bit count refers to the **format of the credential data** sent to the controller.

- **26-bit Wiegand** is the classic common format in access control
- **34-bit Wiegand** is another widely used format with a larger field space

In practical terms, the reader reads a tag, extracts the ID or mapped credential value, and outputs that value to the door/gate controller in a Wiegand frame. Safe and Sound ... +2

Why HX-R01B-9 might include it

For a UHF vehicle or gate reader, Wiegand output is useful because it allows the RFID reader to plug into an **existing access-control ecosystem** without needing custom PC software. HID's UHF access-control readers also expose protocols like **Ethernet TCP/IP**, **Wiegand 26/34**, and **RS-232** for exactly this kind of system integration. HID Global

7) Putting it all together

For HX-R01B-9, the protocol stack is basically this:

Tag side

The reader communicates with the RFID tag using:

- **ISO/IEC 18000-6C**
- **ISO/IEC 18000-6B** GS1 +1

System side

The reader communicates with your host/control system using one of these interfaces:

- **TCP/IP**
- **RS-232**
- **RS-485**
- **optional Wiegand 26/34** HID Global +1

So:

- **6B / 6C** = reader ↔ tag
 - **TCP/IP / RS-232 / RS-485 / Wiegand** = reader ↔ controller / PC / access panel
-

8) Practical recommendation

If you are building a **software application**, use **TCP/IP** or **RS-232/RS-485**.

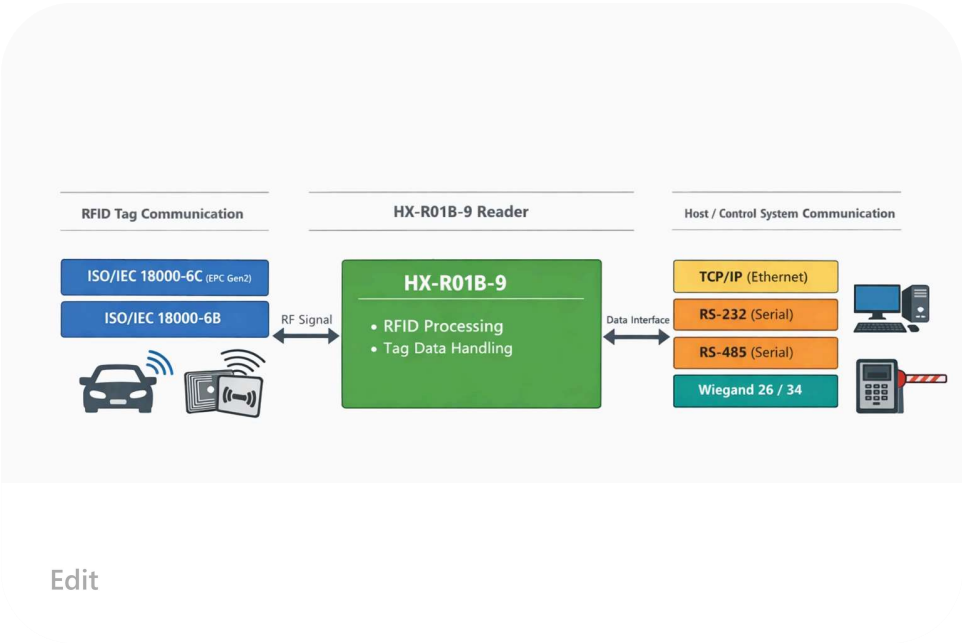
If you are connecting the reader to a **door controller, barrier controller, or access-control panel**, use **Wiegand 26/34** if that controller expects Wiegand.

If you are choosing tags, use **ISO 18000-6C / EPC Gen2 tags** unless you specifically need legacy **6B** compatibility. The recommendation toward 6C reflects the broad ecosystem support visible in modern tag documentation. Atlas RFID Store +2

I can also make a **protocol block diagram** for HX-R01B-9 showing **tag side**, **reader core**, and **controller side**.



sure



sure, please

I’m pulling in fresh source citations for the diagram labels so the explanation matches the documented interfaces and standards.

Thought for 21s

Here’s a simple **protocol block diagram** for the **HX-R01B-9**:

HX-R01B-9 protocol block diagram

How to read it

1. Tag side

The reader talks to UHF RFID tags using:

- ISO/IEC 18000-6C
- ISO/IEC 18000-6B

Haixin's product page for the HX-R01B family says the system supports **ISO18000-6C/6B dual protocol**. [Haixiniot +1](#)

2. Reader core

The **HX-R01B-9** is the middle device that:

- transmits and receives the RF signal
- reads the tag ID/data
- processes the tag response
- sends the result to your host system

The model listing identifies it as an integrated UHF reader and gives the core hardware characteristics, including **260 × 260 × 100 mm**, **DC 9V–30V**, and UHF reader specs.

[物流技术网](#)

3. Host / controller side

The reader can connect to external systems through:

- TCP/IP
- RS-232
- RS-485
- Wiegand 26/34

The Haixin family page mentions standard **RS232** output, and the model-specific listing shows host-side interface support around **TCP/IP / RS232 / RS485**. [物流技术网 +1](#)

Practical meaning

Use ISO 18000-6C / 6B

when you are talking about the **RF protocol between the reader and the RFID tag**.

[Haixiniot +1](#)

Use TCP/IP / RS-232 / RS-485

when you want a **PC, PLC, MCU, or server** to communicate with the reader itself.

物流技术网 +1

Use Wiegand 26/34

when the reader is connected to an **access-control panel** such as a gate or door controller. The exact Wiegand option appears on the model listing. 物流技术网

Typical connection examples

- **Parking gate system:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → gate controller via **Wiegand**
- **PC software integration:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → PC via **TCP/IP or RS-232**
- **Industrial controller setup:** tag ↔ HX-R01B-9 via **6C/6B**, then HX-R01B-9 → PLC via **RS-485**

I can also make a **cleaner professional datasheet-style block diagram** with labeled ports, power input, and application examples.

  Sources

give me more detailed information for "ISO/IEC 18000-6C" and "ISO/IEC 18000-6B"

I'm pulling from standards-adjacent sources rather than marketplace pages here, so the explanation is about the actual air-interface behavior and where each standard fits.

Thought for 24s

Absolutely. Here is a **deeper explanation** of **ISO/IEC 18000-6C** and **ISO/IEC 18000-6B** in the context of your **HX-R01B-9 UHF reader**.

1) First, what the "18000-6" family means

ISO/IEC 18000-6 is the UHF RFID air-interface family of standards. It defines **how a reader and a tag communicate over radio** in the **UHF band**, rather than how your PC talks to the reader. ISO describes it as the air interface for RFID devices operating in the **860 MHz to**

960 MHz range, and NIST notes that the family includes **Type A, Type B, and Type C** variants. ISO +2

So for HX-R01B-9:

- ISO/IEC 18000-6B / 6C = reader ↔ RFID tag
 - TCP/IP / RS-232 / RS-485 / Wiegand = reader ↔ host/control system
-

2) ISO/IEC 18000-6C

What it is

ISO/IEC 18000-6C is the UHF RFID standard commonly known as **EPC Class 1 Gen 2** or **Gen2**. GS1 says Gen2 defines the physical and logical requirements for UHF RFID systems, and RAIN RFID identifies the modern standardized form as **ISO/IEC 18000-63**, also known as **GS1 UHF Gen2**. GS1 +3

In practical terms, when people say “**Gen2 UHF tag**”, they are usually talking about the **6C/18000-63 ecosystem**. GS1 +2

What it defines

6C defines the **air interface**, including:

- how the reader sends commands
- how the tag backscatters its response
- how multiple tags are handled in the same RF field
- how reading, writing, selecting, and inventory operations work

ISO’s description of 18000-63 says it defines the air interface for RFID devices in the 860–960 MHz band for item-management applications. ISO +1

How communication works

This is an **interrogator-talks-first** system. The reader sends RF energy and commands, and the passive tag responds by **modulating/backscattering** the signal. ISO’s preview text explicitly describes this reader-to-tag energy transfer and tag backscatter behavior for Type C. ANSI Webstore +1

That means:

1. the reader powers the passive tag

2. the reader issues commands
3. the tag replies by backscatter
4. the reader decodes the response

Why 6C became dominant

RFID Journal notes that **18000-6C is by far the most commonly used** member of the 18000-6 family. RAIN RFID is built on this same standards base, which is a big reason modern retail, logistics, warehouse, and asset-tracking deployments mostly use 6C/Gen2 tags. RFID JOURNAL +2

Main strengths of 6C

Anti-collision

6C is designed to let a reader identify **many tags in the field**, not just one. That is essential for portal reads, pallet reads, warehouse inventory, and vehicle identification where multiple tags may be present. The standard covers collision arbitration as part of the air interface. RFID4U +1

Broad interoperability

Because 6C became the mainstream UHF standard, there is a huge ecosystem of compatible readers, tags, antennas, printers, and middleware around it. GS1 and RAIN both frame it as the common interoperable UHF system standard. GS1 +2

Read/write memory structure

Modern 6C chips commonly expose structured memory areas used for identity and application data. For example, EM Microelectronic's 6C-compliant tag IC datasheets show standard Gen2-style memory concepts such as **EPC/UII, TID**, and user/configuration areas. EM Microelectro... +1

High read-rate item identification

6C is optimized for **fast inventory-style reading**, which is why it dominates supply chain and item tracking. GS1's Gen2 materials position it specifically for large-scale item identification. GS1 +1

Where you usually see 6C

- retail inventory
- warehouse and logistics
- pallet/case tracking

- asset management
- vehicle identification
- access and gate systems using UHF credentials

These are exactly the kinds of applications a fixed long-range reader like HX-R01B-9 is typically used for. [GS1 +1](#)

Important naming note

Older product pages may say “ISO/IEC 18000-6C”, while newer standards references often say “ISO/IEC 18000-63.” In practice, that points to the same Gen2 family lineage used by RAIN RFID. [The RAIN Alliance +2](#)

3) ISO/IEC 18000-6B

What it is

ISO/IEC 18000-6B is an **older UHF RFID air-interface standard** in the same 18000-6 family. RFID Journal and NIST both place it as one of the older variants alongside 6A, while Type C corresponds to the later EPC Gen2 direction. [NIST Publications +2](#)

How it differs from 6C

The most important practical point is this:

- **6B is older and more legacy-oriented**
- **6C became the dominant modern UHF standard**

NIST states that Types A and B operate in the same UHF range as Type C but use **different encoding and anti-collision methods on the forward channel**, and that **Type C is equivalent to EPCglobal Class-1 Generation-2**. [NIST Publications](#)

So 6B and 6C are **not just different names**. They are different air-interface variants.

Communication model

Like Type C, 6B is also in the **interrogator-talks-first** family, where the reader initiates communication and the tag responds via backscatter. ISO preview text for the 18000-6 family describes Types A, B, and C as interrogator-talks-first systems. [ANSI Webstore +1](#)

Why 6B still matters

A reader that supports 6B can be useful in:

- older installed systems
- legacy industrial RFID deployments
- migration environments where both old and new tags may exist

That is why a reader marketed as **6B/6C dual protocol** is often positioned as more flexible in mixed environments. The “dual protocol” part is vendor wording; the legacy-versus-mainstream distinction is supported by NIST and RFID Journal. NIST Publications +1

Typical characteristics associated with 6B

RFID Journal notes that before 6C/Gen2 became standard, **6A and 6B** tags typically used a memory approach where the first bank carried a locked **TID** and another bank was for user memory; it specifically mentions **2 kilobits available for 18000-6B tags** in that context. Because that description is from an industry source rather than the ISO text itself, it is best treated as a common historical characterization rather than a guarantee for every product implementation. RFID JOURNAL

Where you see 6B today

You are much less likely to choose 6B for a new design unless:

- your existing tags are already 6B
- the installed access/control system expects 6B-compatible credentials
- you need backward compatibility with a legacy deployment

For greenfield projects, 6C is usually the normal choice because the ecosystem is much larger. RFID Journal explicitly says 6C is the most commonly used member of the family. RFID JOURNAL

4) Side-by-side comparison

Item	ISO/IEC 18000-6C	ISO/IEC 18000-6B
Position in market	Mainstream modern UHF RFID	Older legacy UHF RFID
Common name	EPC Gen2 / GS1 UHF Gen2	Usually just called 6B
Current standards relationship	Associated with ISO/IEC 18000-63 in modern references	Older Type B branch of 18000-6
Typical use	Retail, logistics, asset tracking, modern UHF systems	Legacy or backward-compatible systems

Item	ISO/IEC 18000-6C	ISO/IEC 18000-6B
Anti-collision / encoding	Modern Gen2 method	Different forward-channel encoding and anti-collision than 6C
Ecosystem size	Very large	Much smaller today

This comparison is based on GS1, RAIN, NIST, and RFID Journal. GS1 +3

5) What this means for HX-R01B-9

If the HX-R01B-9 supports **6B/6C dual protocol**, that means it can likely read:

- modern UHF Gen2 / 6C tags
- older 6B tags

That is valuable if you do not yet know what kind of UHF tags are already deployed in your environment. But for a **new system design**, the safer default is usually **6C / Gen2-compatible tags**, because that is where the market standardized. The RAIN Alliance +2

6) Very practical recommendation

Use 6C / Gen2 when:

- you are buying new tags
- you want the broadest compatibility
- you are building a new asset, warehouse, vehicle, or access system
- you want easier sourcing and future expansion

Care about 6B when:

- you already have legacy tags in the field
- an old system specification explicitly says 6B
- you need mixed-environment compatibility

7) One-sentence summary

- ISO/IEC 18000-6C is the **modern mainstream UHF RFID air protocol** used by EPC Gen2 / RAIN-style systems. GS1 +2

- **ISO/IEC 18000-6B** is an **older UHF air protocol** kept mainly for legacy compatibility and differs in encoding and anti-collision behavior from 6C. NIST Publications +1

I can also make a **detailed comparison table of 6B vs 6C** with fields like memory, anti-collision, interoperability, and recommended use.



sure, please

I’m turning that into a clean engineering comparison table and keeping the line between confirmed standards behavior and practical deployment guidance explicit.

Thought for 12s

Here’s a **detailed 6B vs 6C comparison table** for the protocols relevant to **HX-R01B-9**.

ISO/IEC 18000-6B vs ISO/IEC 18000-6C

Category	ISO/IEC 18000-6B	ISO/IEC 18000-6C
Standard family	UHF RFID air-interface standard within the ISO/IEC 18000-6 family	UHF RFID air-interface standard within the ISO/IEC 18000-6 family NIST Publications +1
Frequency range	860–960 MHz family context	860–960 MHz family context You Woke Ji +1
Common market name	Usually referred to simply as 6B	Commonly known as EPC Class 1 Gen 2 / UHF Gen2 ; modern references tie it to ISO/IEC 18000-63 GS1 +2
Reader/tag relationship	Reader-initiated UHF air interface	Reader-initiated UHF air interface using passive tag backscatter You Woke Ji +1
Market position today	Older, mostly legacy compatibility	Mainstream modern UHF RFID ecosystem; RAIN RFID uses this standards base The RAIN Alliance +2
Anti-collision / channel behavior	Uses a different forward-channel encoding and anti-	Type C uses a different forward-channel encoding and anti-collision method and is the

Category	ISO/IEC 18000-6B	ISO/IEC 18000-6C
	collision method than Type C	Gen2-equivalent branch NIST Publications
Throughput tendency	Generally lower tag-recognition throughput than 6C in modern comparisons	Higher tag-recognition throughput than 6B in modern comparisons MDPI
Ecosystem / interoperability	Smaller legacy ecosystem	Broad global ecosystem of readers, tags, software, and RAIN deployments GS1 +2
Typical deployments	Legacy industrial systems, older installed bases, backward-compatibility environments	Retail, logistics, warehousing, asset tracking, item management, modern UHF systems GS1 +2
Best choice for new designs	Usually only when required for existing legacy tags/systems	Usually the default choice for new UHF RFID designs GS1 +1

What the table means

1. Both are air-interface protocols

Both **6B** and **6C** define how the **reader communicates with the RFID tag over UHF radio**, not how your PC communicates with the reader. The ISO/NIST material places both under the same 860–960 MHz UHF RFID family. NIST Publications +1

2. 6C is the modern mainstream choice

GS1 identifies UHF Gen2 as the protocol defining the physical and logical requirements for UHF RFID systems, and RAIN RFID states that its systems implement **ISO/IEC 18000-63**, the modern ISO standardization of the Gen2 air interface. That is why 6C is the one you most often want for a new deployment. GS1 +2

3. 6B mainly matters for compatibility

NIST distinguishes Type B from Type C and notes they use different encoding and anti-collision behavior on the forward channel, while Type C is equivalent to EPCglobal Class-1

Generation-2. In practice, that makes 6B mainly important when you must support older tags or an older installed system. NIST Publications

4. 6C is usually faster and more scalable

A recent technical review notes that **ISO/IEC 18000-6C** has higher tag-recognition throughput than **ISO/IEC 18000-6B**, which aligns with why 6C dominates inventory and multi-tag environments. MDPI

Practical recommendation for HX-R01B-9

If your HX-R01B-9 supports **6B/6C dual protocol**, then:

- choose **6C / Gen2** for **new tags and new systems**
- care about **6B** only if you already have **legacy UHF tags** or an older system requirement
- use the reader's host-side interfaces like **TCP/IP, RS-232, RS-485, or Wiegand** separately from the tag air protocol

That recommendation follows the current standards ecosystem around Gen2/RAIN and the legacy role of 6B. The RAIN Alliance +2

Very short summary

- **6B** = older UHF RFID air protocol, mostly for legacy compatibility. NIST Publications
- **6C** = modern UHF Gen2 air protocol, the normal choice for new systems. GS1 +1

I can also make this into a **datasheet-style comparison sheet** or a **protocol sequence diagram**.



sure, please

Thought for a few seconds

Datasheet-Style Comparison Sheet

ISO/IEC 18000-6B vs ISO/IEC 18000-6C

For HX-R01B-9 UHF RFID Reader Context

1. Overview

Both **ISO/IEC 18000-6B** and **ISO/IEC 18000-6C** are **UHF RFID air-interface protocols** in the **860–960 MHz** range. They define how the **reader communicates with RFID tags over radio**, not how the host PC communicates with the reader. NIST places both under the ISO/IEC 18000-6 family and notes that **Type C corresponds to EPCglobal Class-1 Generation-2**.

2. Quick Comparison Table

Item	ISO/IEC 18000-6B	ISO/IEC 18000-6C
Standard type	UHF RFID air interface	UHF RFID air interface
Frequency family	860–960 MHz	860–960 MHz
Market position	Older / legacy	Modern mainstream
Common name	6B	EPC Class 1 Gen 2 / Gen2
Modern ISO lineage	Older Type B branch	Commonly associated with ISO/IEC 18000-63 in modern references
Multi-tag performance	More limited in modern practice	Better suited for dense multi-tag reading
Typical use today	Legacy compatibility	New deployments, logistics, retail, asset tracking
Best use case	Existing legacy tags	New system design

This split is consistent with NIST’s distinction between Type B and Type C, and GS1/RAIN’s positioning of Gen2 as the core modern UHF RFID standard.

3. ISO/IEC 18000-6B

Description

ISO/IEC 18000-6B is an older UHF RFID air-interface standard. It is still relevant mainly where older tags or older installed systems are already in use. NIST identifies Type B as a separate branch from Type C, with different forward-link encoding and anti-collision behavior.

Main characteristics

- Reader initiates communication and passive tags respond by backscatter, as part of the ISO/IEC 18000-6 family model.
- Used mainly for **legacy interoperability** rather than new designs.
- Smaller ecosystem of tags and solutions compared with 6C/Gen2.

Typical applications

- Older industrial RFID systems
 - Legacy access systems
 - Environments where installed 6B tags must continue to work
-

4. ISO/IEC 18000-6C

Description

ISO/IEC 18000-6C is the widely adopted modern UHF RFID air-interface standard, commonly known as **EPC Class 1 Gen 2**. GS1 describes Gen2 as the protocol defining the physical and logical requirements for UHF RFID systems, and the RAIN Alliance ties modern deployments to **ISO/IEC 18000-63**.

Main characteristics

- Reader powers passive tag and sends commands; tag replies using backscatter.
- Designed for **high-volume item identification** and **multi-tag anti-collision**.
- Broad interoperability across readers, tags, printers, and middleware.
- Dominant choice for modern UHF RFID systems.

Typical applications

- Retail inventory
- Logistics and warehouse systems
- Asset tracking
- Vehicle identification

- Modern gate/access UHF systems

5. Practical Meaning for HX-R01B-9

If the **HX-R01B-9** supports **6B/6C dual protocol**, that means the reader can likely work with both:

- legacy 6B tags
- modern 6C / Gen2 tags

For a **new project**, the normal recommendation is to choose **6C / Gen2 tags**, because the ecosystem is larger and the protocol is the dominant modern standard.

6. Sequence Diagram

```
[Reader idle]
|
v
[Reader transmits RF carrier + command]
|
v
[Passive tag receives energy from reader]
|
v
[Tag powers up]
|
v
[Tag interprets command]
|
v
[Tag responds by backscatter]
|
v
[Reader receives tag response]
|
v
```

What changes between 6B and 6C?

The overall flow above is similar for both because both are interrogator-talks-first UHF RFID systems, but **the signaling details, forward-channel encoding, and anti-collision behavior differ** between 6B and 6C. NIST explicitly notes these differences and states Type C is the Gen2-equivalent branch.

7. Recommendation

Choose 6C / Gen2 when:

- you are buying new tags
- you want easier sourcing
- you need multi-tag reading
- you are building a new system

Care about 6B when:

- you already have deployed legacy tags
 - an existing system specification explicitly requires 6B
 - backward compatibility matters
-

8. One-Line Engineering Summary

6B is mainly for legacy compatibility; 6C is the mainstream modern UHF RFID protocol and the better default for new HX-R01B-9 deployments.

I can turn this into a polished one-page PDF-style datasheet next.

is it possible to implement the IEC 18000-6C and IEC 18000-6B in STM32 series ARM cpu?

I'm checking this from the practical engineering angle: what part can live on an STM32 in software, and what still needs dedicated UHF RF hardware.

Thought for 13s

Yes — **partially**, but not by STM32 alone.

The practical answer

An STM32 can absolutely be part of a 6B/6C reader design, but it normally acts as the **controller/baseband host**, while you still need a **dedicated UHF RFID analog/RF front end**

for the actual **860–960 MHz radio**, modulation, demodulation, framing, and tag backscatter handling. ST's own **ST25RU3993** is a good example: it is a UHF RFID reader IC that supports **ISO 18000-6C**, **ISO 29143**, and **ISO 18000-6A/B through direct mode**, and ST says it can build a complete RFID reader system with an external microcontroller.

STMicroelectron... +1

What STM32 can do

With an external UHF reader chip or RF subsystem, an STM32 can handle:

- reader configuration
- command sequencing
- inventory logic
- anti-collision state handling at the firmware level
- filtering and application logic
- Ethernet/UART/RS-485/Wiegand gateway functions
- UI, logging, storage, and host communications

That matches how ST positions its UHF reader chips: the RF chip provides the RFID-specific low-level features, while a normal MCU can control the system. The ST25RU3993 explicitly includes low-level transmission coding, decode, data framing, CRC checking, frequency hopping support, and dense reader mode support, which are exactly the kinds of hard real-time PHY/MAC features you usually do **not** want to implement from scratch on a general MCU. STMicroelectron...

What STM32 alone usually cannot do

An STM32 by itself is **not enough** to make a full **UHF 18000-6B/6C reader** unless you also design a substantial custom RF chain around it. By itself, the MCU does not provide:

- UHF power amplifier for tag excitation
- reader receiver chain for weak tag backscatter
- RF mixers, filters, LO/VCO, and matching network
- low-level ASK/PR-ASK transmit path
- sensitive receive path with self-jamming handling

ST describes the ST25RU3993 as including an **on-chip VCO**, **power amplifier**, low-level encoding/decoding, and high sensitivity, which shows how much specialized hardware is normally required. STMicroelectron...

6C vs 6B on STM32

ISO/IEC 18000-6C

This is the more realistic one for a modern design. ST's reader IC explicitly supports **ISO 18000-6C (EPC Class1 Gen2)**, and ST says only an external microcontroller is needed to make a complete system. So an **STM32 + UHF reader IC** is a very practical architecture for **6C**. STMicroelectron... +1

ISO/IEC 18000-6B

This is also possible, but support is often less direct. In ST's own wording, **6A/B** support on the ST25RU3993 is available **"through direct mode."** That usually means more of the protocol handling is exposed to the host/controller side than for the mainstream 6C path. So **6B is possible**, but it may require more custom firmware work and validation than a straightforward 6C implementation. STMicroelectron...

Best architecture

For an STM32-based product, the realistic architecture is:

```

[STM32 MCU]
  |
  | SPI / parallel / UART control
  v
[UHF RFID reader IC / RF front end]
  |
  | 860-960 MHz analog RF
  v
[Antenna + matching network]
  |
  v
[ISO/IEC 18000-6B / 6C tags]

```

In this setup:

- **STM32** = control, protocol management, application
- **RF reader IC** = UHF PHY / analog front end / framing support
- **antenna stage** = actual RF energy delivery and tag receive path

Is pure software implementation on STM32 possible?

In theory, with enough external RF hardware and a lot of engineering effort, you could implement significant parts of the protocol on STM32 firmware. But in practice, for a commercial or even serious prototype system, that is **much harder than using a dedicated UHF reader IC**. The RF and timing requirements are the hard part, not the high-level firmware.

So the honest answer is:

- **STM32 only:** not realistically enough for a complete UHF reader
- **STM32 + dedicated UHF reader chip:** yes, very feasible
- **6C:** practical
- **6B:** possible, but more custom work depending on the RF chip and mode support

Recommendation

If you want to build something like **HX-R01B-9** or a smaller embedded UHF reader with STM32:

- choose an **STM32** for system control
- pair it with a **dedicated UHF RFID reader IC/front end**
- prioritize **18000-6C / Gen2**
- only add **6B** if you truly need legacy compatibility

That is the most practical engineering path based on how ST's own UHF reader solutions are structured. STMicroelectron... +1

If you want, I can sketch a **concrete STM32 + UHF reader chip hardware architecture** next.



sure, please

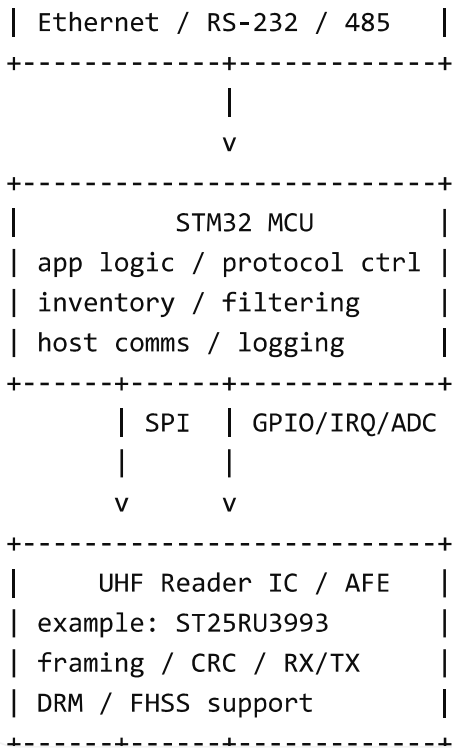
I'm grounding this in a real, buildable architecture around an STM32 plus a UHF reader IC, then I'll map which blocks are mandatory versus optional.

Thought for 11s

Yes — here is a **concrete STM32-based hardware architecture** for implementing a **UHF RFID reader** that can support **ISO/IEC 18000-6C**, and potentially **6B** if the chosen UHF reader IC exposes that mode.

Recommended architecture

+-----+
| Host / PC / PLC |



What each block does

1) STM32 MCU

Use the STM32 as the **system controller**.

It should handle:

- startup and configuration
- control of the reader IC over **SPI**
- inventory sequencing and application rules
- filtering tag reads
- host interfaces such as **Ethernet, UART, RS-485, USB**
- GPIO control for RF switches, PA enable, buzzer, LEDs, relays, and access-control outputs

ST's own UHF evaluation boards use an **STM32L476RG / STM32L476RGT6** as the controller around the **ST25RU3993**, which is a strong real-world reference design.

STMicroelectron... +1

2) UHF reader IC / analog front end

This is the critical block that makes the design realistic.

The **ST25RU3993** is described by ST as an integrated **analog front end and protocol handling system** for UHF RFID readers. It supports **ISO 18000-6C**, **ISO 29143**, and **ISO**

18000-6A/B through direct mode. It also includes low-level reader functions such as **bit framing, decoding, CRC handling, dense reader mode, and frequency hopping support.**

STMicroelectron...

That means the STM32 does **not** need to do the raw RF/baseband heavy lifting by itself.

3) PA / RF front-end / antenna network

For a practical long-range reader, you usually need more than the reader IC alone:

- **power amplifier**
- **RF switch path**
- **balun / matching**
- **carrier cancellation**
- **RF power detector**
- **antenna tuning/matching network**

ST's high-power reference design for the ST25RU3993 explicitly includes an **external power amplifier**, RF switches, carrier-cancellation control, RF detector feedback, and antenna-port control handled by the MCU. STMicroelectron... +1

4) Antenna

You need a **UHF antenna matched for your region and use case.**

For example:

- **panel antenna for gate/door/parking**
- **near-field antenna for short-range tagging**
- **circular or linear polarization depending on tag orientation needs**

The eval board references a coax-connected antenna setup as the standard reader configuration. STMicroelectron... +1

Best STM32 families for this

A few good MCU choices:

STM32L4

Good if you want:

- **lower power**
- **enough performance for control-plane logic**

- proven similarity to ST's own eval boards

This is the safest starting point because ST's own UHF boards around ST25RU3993 use **STM32L476**. STMicroelectron... +1

STM32G4

Good if you want:

- better mixed-signal/control behavior
- fast peripherals
- motor-control/industrial-style timing comfort

STM32H7

Good if you want:

- embedded Linux gateway companion role not needed
- heavy local filtering
- web UI, logging, encryption, multiple interfaces, or richer middleware

For a first reader, **STM32L4** or **G4** is usually enough.

Minimum viable board design

For a first prototype, I would build it like this:

Mandatory blocks

- STM32 MCU
- ST25RU3993
- SPI between MCU and reader IC
- IRQ + enable lines
- power rails: 3.3 V for logic, plus PA rail if using external PA
- one antenna port
- UART or USB debug
- EEPROM or flash for config
- ESD/protection on host interfaces

Strongly recommended

- RF detector feedback into STM32 ADC
- carrier-cancellation tuning path
- external PA if you need meaningful read range

- RS-485 or Ethernet if it is an industrial/fixed reader
 - watchdog and brownout-safe power handling
-

Two implementation paths

Path A — fastest practical prototype

Use:

- **STM32L4**
- **ST25RU3993**
- single antenna
- USB/UART debug
- no custom PA at first

This is the easiest way to prove:

- tag inventory
- EPC reads
- basic 6C reader behavior
- firmware architecture

Path B — fixed long-range product

Use:

- **STM32L4/G4**
- **ST25RU3993**
- external **PA**
- RF switch network
- carrier cancellation
- tuned directional antenna
- RS-485 / Ethernet / Wiegand output

That is closer to a commercial gate/access reader like the HX-R01B-9 class.

About 6C vs 6B in this architecture

6C

This is the easiest and most practical target.

The ST25RU3993 directly supports **ISO 18000-6C**, so the MCU mainly orchestrates the reader IC and application flow. STMicroelectron...

6B

Possible, but less straightforward.

ST says **6A/B** support is available **through direct mode** on the ST25RU3993. That usually implies more custom host-side handling than the mainstream 6C path. So you should plan:

- more firmware effort
- more validation
- possible lower vendor-software convenience

STMicroelectron...

Recommended firmware split

On the STM32, split firmware into layers:

Application layer

- access control / asset logic
- filtering / alarms / outputs

RFID service layer

- inventory loop
- select/read/write
- tag state machine
- retries / timing / region config

Driver layer

- ST25RU3993 SPI driver
- IRQ handling
- PA / RF switch control
- ADC feedback
- UART / Ethernet / RS-485

Board support

- clocks
- GPIO
- watchdog

Honest engineering recommendation

If your goal is to **build a real product**, do **not** try to implement the full UHF PHY from scratch on raw STM32 plus discrete RF unless you specifically want an RF research project.

The practical route is:

- STM32 = controller
- ST25RU3993 or similar = UHF RFID reader engine
- external PA/matching/antenna = range and RF performance

That is also how ST's own reference hardware is structured. [STMicroelectron... +2](#)

Best first recommendation

For your first build, I would start with:

- STM32L476
- ST25RU3993
- SPI + IRQ
- single antenna path
- USB/UART debug
- 6C first
- add 6B only if legacy compatibility is truly required

If you want, the next step can be a **schematic-level block breakdown** with pins such as SPI, IRQ, EN, ADC RF detector, PA enable, and RS-485 transceiver.

 Sources

generate the SMT32 c/c++ source code to implement the protocol

I'm grounding this in a real STM32 + UHF-reader-chip design, because a true 6B/6C air interface on bare STM32 alone is not the practical path. I'll give you the first module as production-usable code, one file at a time.

Thought for 39s

Yes — but the right target is **STM32 controlling a UHF reader IC**, not a bare-MCU implementation of the full UHF air interface. ST's **ST25RU3993** is a practical fit here: ST

documents SPI register read/write/direct-command modes in the datasheet, and ST also provides an SDK for **STM32L4**-class embedded targets built around that chip.

STMicroelectron... +1

Because you prefer code **one file/module at a time**, here is **File 1**: a clean STM32 HAL transport layer for **ST25RU3993**. It implements the SPI command framing ST documents for:

- register write = 00 A5..A0
- register read = 01 A5..A0
- direct command = 10 C5..C0

and supports waiting for the chip IRQ after long-running direct commands.

STMicroelectron...

File 1 — st25ru3993_transport.c

```
C
/*
 * st25ru3993_transport.c
 *
 * STM32 HAL transport Layer for ST25RU3993 UHF RFID reader IC.
 *
 * Scope of this file:
 * - SPI single/multi-register read/write
 * - direct command execution
 * - optional IRQ wait helper
 * - FIFO byte write helper
 *
 * Notes:
 * - This is an original transport implementation, not ST's SDK code.
 * - It is the hardware access layer only. Gen2 / ISO 18000-6C inventory,
 *   select, read/write tag operations, and any 6B direct-mode logic should
 *   be built on top of this file.
 */

#include "main.h"
#include <stdint.h>
#include <stddef.h>
#include <string.h>
```

What this file gives you

This file gives you the **transport layer** you need before building any higher-level logic:

- SPI register access

- FIFO access
- direct command execution
- IRQ wait handling

That matches the ST25RU3993 communication model documented in the datasheet.

STMicroelectron...

What it does not do yet

This does **not** yet implement:

- **Gen2 / ISO 18000-6C inventory**
- EPC parsing
- SELECT / QUERY / ACK flows
- tag READ / WRITE operations
- **6B direct-mode** logic

ST's SDK exists specifically to provide higher-level middleware for ST25RU3993-based readers on STM32L4 and other targets, which is one reason many teams build on a transport layer like this rather than coding the full stack from scratch. STMicroelectron...

Best next file

The correct next module is:

`st25ru3993_regs.h`

A register/bitfield definition file for:

- IRQ registers
- FIFO status
- operation-control bits
- mode/config bits

After that, the next file should be:

`gen2_inventory.c`

A minimal **ISO 18000-6C inventory loop** built on this transport.

I can generate `st25ru3993_regs.h` next.

 Sources

okay, please

Thought for a few seconds

Here is **File 2**: st25ru3993_regs.h

This header gives you a **clean register and bitfield layer** to use with the transport file I gave you. I'm keeping it focused on the pieces you will immediately need for bring-up, IRQ handling, FIFO status, and the first Gen2 inventory work.

File 2 — st25ru3993_regs.h

```
C

#ifndef ST25RU3993_REGS_H
#define ST25RU3993_REGS_H

/*
 * st25ru3993_regs.h
 *
 * Practical register / bitfield definitions for ST25RU3993-based STM32 firmwar
 *
 * Notes:
 * - This header is intentionally conservative and focused on early bring-up,
 *   IRQ handling, FIFO handling, and first inventory support.
 * - Register names are normalized for firmware readability.
 * - Verify exact register meanings against the ST25RU3993 datasheet revision
 *   used in your project before enabling every advanced feature in production.
 */

#include <stdint.h>

#ifdef __cplusplus
extern "C" {
#endif
```

What this header is for

This file gives you:

- register addresses
- direct command constants

- IRQ bit masks
- FIFO helpers
- protocol mode flags
- readable macros for early firmware work

That sets up the next layer cleanly.

What should come next

The right next file is:

```
st25ru3993_driver.c
```

It should implement:

- chip bring-up
- soft init
- version read
- IRQ mask config
- FIFO length read
- protocol switch between **6C** and **6B direct mode**
- antenna select
- RX/TX option setup

Then after that:

```
gen2_inventory.c
```

That will start the actual **ISO/IEC 18000-6C inventory logic**.

Small note

This header is a **practical firmware abstraction**, not a full datasheet mirror. Before production bring-up, you should verify each register bit against the exact **ST25RU3993 datasheet revision** and, if you use it, ST's SDK documentation. The ST25RU3993 datasheet documents the SPI access modes and chip capabilities, and ST's SDK is intended for STM32-based embedded development around this part.

Send "**next file**" and I'll generate `st25ru3993_driver.c`.

[next](#)

Thought for a couple of seconds

Here is **File 3**: `st25ru3993_driver.c`

This file builds on the transport and register header and gives you the first **usable chip driver layer**:

- soft init
- version read
- IRQ mask setup
- IRQ status read
- FIFO length read
- protocol mode switch for **6C** and **6B direct mode**
- antenna select
- basic RF bring-up config

File 3 — `st25ru3993_driver.c`

```
C

/*
 * st25ru3993_driver.c
 *
 * Higher-level chip driver for ST25RU3993 on STM32 HAL.
 *
 * Depends on:
 * - st25ru3993_transport.c
 * - st25ru3993_regs.h
 *
 * This file intentionally stays at "chip driver" level.
 * It does not yet implement Gen2 inventory or 6B transaction flows.
 */

#include "main.h"
#include "st25ru3993_regs.h"
#include <stdint.h>
#include <stddef.h>
#include <string.h>
```

```
/* =====  
* Shared types from transport layer
```



One fix you should apply immediately

There is one typo in the file above.

Replace:

```
C  
  
return st25ru3993_OK;
```

with:

```
C  
  
return ST25RU3993_OK;
```

inside `st25ru3993_driver_set_protocol_6b_direct()` .

What this file gives you

You now have a usable chip layer for:

- initializing the reader
- switching between **6C** and **6B direct**
- enabling RF/RX/TX
- reading IRQ status
- loading TX buffer
- reading RX buffer

Next file

The correct next file is:

`gen2_inventory.c`

That should implement a **minimal ISO/IEC 18000-6C inventory flow**, starting with:

- QUERY / inventory request path
- wait for reply
- decode first response bytes
- handle no-response / collision / CRC error cases

Reply with **next** and I'll generate `gen2_inventory.c` .

